



Retry Exam (28 Feb. 2024)

Begonnen am	Mittwoch, 28. Februar 2024, 15:41
Status	Beendet
Beendet am	Mittwoch, 28. Februar 2024, 17:08
Verbrauchte Zeit	1 Stunde 27 Minuten
Bewertung	94,33 von 100,00

Frage 1

Vollständig
Nicht bewertet

Frage
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General information:

- Not counting this first "task" (i.e. the statutory declaration below), there are 24 problems with 100 points in total.
- Read all problems and possible answers carefully and take your time to think about them.
- Multiple-choice questions: Choosing wrong answers leads to deduction of points within that question. However, you cannot get less than 0 points on a question.
- Numerical questions: Wrong answers do not lead to point deductions.
- For numerical answers, use "." for the decimal point. In case you get an error message or the question is marked as not answered, try using "," instead. If that still throws an error, just ignore it. There will be an automatic conversion.
- When it says that you can use a pocket calculator, you may also use the calculator of your operating system.
- After some questions, there is an optional text field. This field will not influence the automatic grading. Here, you can write down subresults of your calculations. It may be used to your advantage at the end of the semester. There will be no other form of submitting information, i.e., no emails or Moodle uploads with photos or pdfs after the exam.
- You can ask questions during the exam only via the chat window and only directly to the exam supervisor. Your multiple-choice answers may be displayed in random order. So, don't refer to them by their label.
- You have 120 minutes for the exam. After this time is expired, your exam will automatically be submitted. There is no overtime.
- Once you have submitted your exam, please state so in the chat window, e.g. write "submitted", don't wait for a reply, and then just leave the Zoom meeting.
- Strictly forbidden: All forms of audio or video recording by the candidates during the examination. This includes, but is not limited to, screenshots, downloads and any copy of the exam content (digital or analogue). Also after the exam, you are not allowed to copy or share or pass on the exam material in any way whatsoever.
- The Zoom session of the exam may be recorded. (By law, recordings must not be kept longer than 72 hours. We will delete it even earlier in case everything runs smoothly, which is what we expect.)

Statutory declaration:

With clicking on "I agree" below, you agree to this statutory declaration ("Eidesstattliche Erklärung"): "I hereby confirm that I complete this closed-book exam on my own and only using the allowed materials, in particular without any written or electronic notes and without any help from or communication with other people. I understand that violation of this declaration may cause legal consequences."

Frage 2

Richtig
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4,00 von 4,00

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Which statements about the generalization error and the empirical risk are true?

Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

- ☒ a. The generalization error is the expected loss on future data for a given model. ✓
- ☐ b. The generalization error has the form $\int_{\mathcal{X}} \int_{\mathcal{Y}} L(y, g(\vec{x}; \vec{w})) p(\vec{x}) p(y) dy d\vec{x}$.
- ☐ c. The empirical risk on a dataset with n samples has the form $R_{\text{emp}} = \frac{1}{n} \prod_{i=1}^n L(y_i, g(\mathbf{x}_i; \mathbf{w}))$.
- ☒ d. A common assumption for estimation of the generalization error via the empirical risk is that data are independent and identically distributed (i.i.d.). ✓
- ☐ e. A good estimation of the generalization error via the empirical risk generally requires that data points are uniformly distributed.
- ☒ f. A good estimation of the generalization error via the empirical risk generally requires a large number of samples. ✓
- ☐ g. A good estimation of the generalization error via the empirical risk generally requires use of a differentiable loss function.
- ☐ h. Estimating the risk via samples from a test set, which have not been used in training, avoids the problem of underfitting.

Frage 3

Richtig
Erreichte Punkte
4,00 von 4,00

Frage
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Which statements about supervised machine learning are true?

Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

- ☒ a. Having a small training error but a large test error indicates overfitting. ✓
- ☐ b. Bias increases with increasing model class complexity.
- ☒ c. ROC curves allow to evaluate a classifier independent of class distribution and independent of misclassification cost. ✓
- ☐ d. Model variance decreases with increasing model class complexity.
- ☒ e. Supervised machine learning uses data with supervisory signals (i.e. labels) for predictive modeling. ✓
- ☐ f. The more degrees of freedom, the lower the risk to fit to noise.
- ☒ g. The Bayes Optimal Classifier is a probabilistic model that makes the most probable prediction for a new sample. ✓
- ☐ h. Supervised machine learning uses explicit knowledge to design models deductively.

Frage 4

Teilweise richtig
Erreichte Punkte
2,00 von 4,00

Markierung
entfernen

Which statements about the probabilistic framework and the Bayes optimal classifier are true?

Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

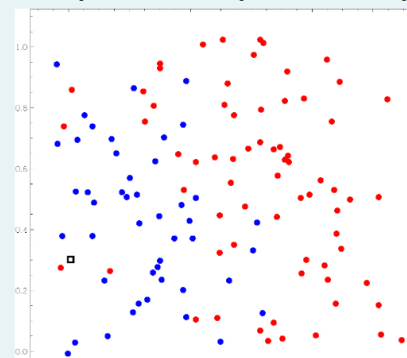
- ☐ a. Using the zero-one loss and binary classification, the minimal risk is given by $\int_{\mathcal{X}} \min[p(y = -1|\mathbf{x}), p(y = +1|\mathbf{x})] p(\mathbf{x}) d\mathbf{x}$
- ☐ b. Bayes Theorem reads: $p(b|a) = [p(a|b)p(a)]/p(b)$.
- ☒ c. The Bayes Optimal Classifier minimizes the misclassification probability, i.e. it minimizes the risk. ✓
- ☐ d. Given two variables, the marginal distribution of one is obtained by integrating (or summing) the conditional distribution over the other variable. If b can take on all real values, then $p(a) = \int_{\mathcal{R}} p(a|b) db$

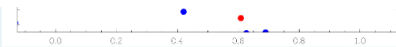
Frage 5

Richtig
Erreichte Punkte
4,00 von 4,00

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markieren

k -nearest neighbors (k -NN): Consider the figure below. There are n training data points in total and more red points ($y = +1$) than blue points ($y = -1$). A test data point $\mathbf{x}_0$ is shown by a black square, and its label is denoted as y_0 .





Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

- ☐ a. For $k = 1$: $y_0 = -1$
- ☒ b. For $k = 3$: $y_0 = -1$ ✓
- ☐ c. For $k = n$: $y_0 = -1$
- ☐ d. $k = 1$ leads to underfitting.
- ☐ e. $k = n$ corresponds to large model complexity.
- ☒ f. In k -NN, the parameter k has to be specified before classification. ✓
- ☒ g. k -NN requires use of a distance measure for measuring closeness. ✓
- ☐ h. k -NN can work with labeled and unlabeled data.

Frage 6

Richtig
Erreichte Punkte
4,00 von 4,00
🚩 Frage markieren

Bias variance decomposition for quadratic loss:

Suppose that labels y for a given feature vector $\mathbf{x}_0$ are normally distributed with mean $\mu = 2$ and standard deviation $\sigma = 0.5$, while the predictions $g(\mathbf{x}_0; \mathbf{w}(\mathbf{Z}_t))$ are normally distributed with mean $\mu = 4$ and variance $\sigma^2 = 2.5$. Compute the expected prediction error for quadratic loss.

You can use a pocket calculator. Give the answer as a real number, rounded to two decimal places. Use "." for the decimal point. Number format example: 1.23

Antwort: ✓

Frage 7

Vollständig
Nicht bewertet
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This text field is optional and will not influence the automatic grading. Here, you can write down subresults or your thoughts on the specific tasks. It may be used to your advantage at the end of the semester.

EPE = $0.5^2 \cdot 2 + (2-4)^2 + 2.5$

Frage 8

Richtig
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5,00 von 5,00
🚩 Frage markieren

In a group of 50000 people, 45000 are healthy, i.e. they are "negative" with respect to a certain disease, and 5000 are "positive", i.e. they have the disease. A medical test is developed. The test has a sensitivity of 60% and a specificity of 80%.

What is the accuracy of the test?

You can use a pocket calculator. Give the answer as a real number between 0 and 1, rounded to two decimal places. Do not give the answer in %. Use "." for the decimal point. Number format example: 0.67

Antwort: ✓

Frage 9

Vollständig
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TP = $5,000 \cdot 0.6 = 3,000$

TN = $45,000 \cdot 0.8 = 36,000$

ACC = $(TP + TN) / (P + N) = 39/50 = 0.78$

Frage 10

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5,00 von 5,00
🚩 Frage markieren

Consider the following classification results of a binary classifier tested on 7 samples:

$\bar{g}(\mathbf{x})$	y
0.1	1
-0.8	-1
1.0	1
0.5	-1
1.9	-1
-0.1	-1
1.4	1

Compute the Area Under Curve (AUC) of the ROC curve.

You can use a pocket calculator. Give the answer as a real number between 0 and 1, rounded to two decimal places. Do not give the answer in %. Use "." for the decimal point. Number format example: 0.67

Antwort: ✓

Frage 11

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sort by $\bar{g}_{\text{bar}}(\mathbf{x})$ in descending order. Plot the curve: x-axis = TNR; y-axis = TPR. Out of twelve "squares", the curve covers 8 -> $8/12 = 2/3 = 0.67$

Frage 12

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🚩 Markierung entfernen

Which statements about support vector machines (SVMs) are true?

Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

- ☐ a. Kernels allow to transform data into a lower-dimensional space where separability can easier be achieved.

- ☒ b. A sample $\mathbf{x}_i$ is a support vector if the corresponding Lagrange parameter α_i is larger than 0. ✓
- ☒ c. SVMs work effectively if the number of dimensions (features) d is much larger than the number of samples l . ✓
- ☐ d. For l data points with features $\mathbf{x}_i$, labels y_i , and Lagrange parameters α_i , the linear SVM classification for a new input $\mathbf{x}$ is given by $g(\mathbf{x}) = \text{sign}(\sum_{i=1}^l \alpha_i \mathbf{x}_i \cdot \mathbf{x} - b)$.
- ☒ e. One of the KKT conditions has the form: $\alpha_i h_i(\mathbf{w}^*) = 0$ for all $i = 1, \dots, l$. ✓
- ☒ f. The mathematical problem (constrained convex optimization) has a global solution. ✓
- ☒ g. Support vector regression uses the magnitude (and not only the sign) of the discriminant function. ✓
- ☐ h. SVMs allow for a probabilistic explanation for classification.

Frage 13

Richtig
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Find the solution $\hat{w}_1, \hat{w}_2$ to the following constrained optimization problem with respect to w_1, w_2 :

$$\begin{aligned} &\text{minimize} && \frac{1}{2}(w_1^2 + w_2^2) \\ &\text{subject to} && 3 - w_1 - 2w_2 \leq 0. \end{aligned}$$

You can use a pocket calculator. Give the answer as a real number, rounded to one decimal place. Use "." or "," for the decimal point (depending on what works with your system). Number format example: 2.1 or 2,1

Below, first enter your solution for $\hat{w}_1$. In the second line, enter the factor by how much you have to multiply $\hat{w}_1$ to get $\hat{w}_2$.

- $\hat{w}_1 =$ ✓
- $\hat{w}_2 =$ ✓ $\hat{w}_1$

Frage 14

Vollständig
Nicht bewertet
Frage markieren

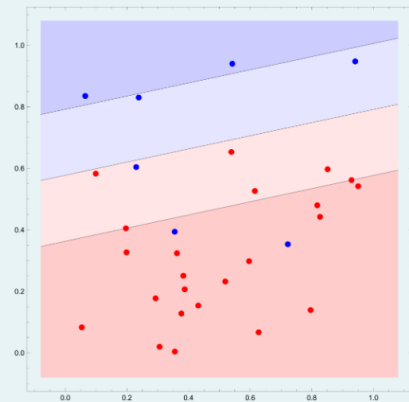
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dL/dw_1 = w_1 - alpha -> w_1 = alpha
dL/dw_2 = w_2 - 2alpha -> w_2 = 2alpha = 2w_1
3 - 5w_1 = 0
w_1 = 3/5 = 0.6
w_2 = 1.2

Frage 15

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Consider a C-SVM classifier trained with a linear kernel on a simple 2D data set (red dots = positive samples, blue dots = negative samples) as depicted below. Blue areas are classified as negative (light blue = inside margin) and red areas are classified as positive (light red = inside margin).



Which of the following statements are correct? Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

- ☒ a. If the Lagrange multiplier α_i is 0 for a data point with index i , then the slack variable ξ_i must be 0. ✓
- ☐ b. Just considering the red data points: Exactly 4 red points are misclassified. They lie inside the margin.
- ☒ c. If the slack variable is larger than 0 for a data point with index i , then also the Lagrange multiplier α_i must be larger than 0. ✓
- ☐ d. Just considering the blue data points: There are exactly 2 blue support vectors. They both lie outside the margin in red territory.
- ☐ e. Considering all data points: Exactly 7 data points require a slack variable larger than 0. They lie inside the margin.

Frage 16

Richtig
Erreichte Punkte
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Frage markieren

Consider multi-class classification with Support Vector Machines. Let there be M classes. Which of the following statements are true? Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

- ☒ a. In the one-versus-one approach, there are $M(M-1)/2$ classifiers. ✓
- ☒ b. In the one-versus-all approach, there are M classifiers. ✓
- ☐ c. In the one-versus-one approach, the final classification is done by choosing the class which produces the largest value of all classifiers.
- ☐ d. In the one-versus-all approach, the final classification is done by majority vote.
- ☐ e. Training pairwise classifiers is usually more costly due to the larger training sets.

Frage 17

Richtig
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Frage markieren

Which statements about decision trees are true?

Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

- ☒ a. Variance reduction is a common splitting criterion for regression. ✓
- ☒ b. Shallow trees tend to underfit, deep trees tend to overfit. ✓
- ☐ c. Decision trees can only be applied to categorical attributes.
- ☒ d. For classification tasks, a new input sample can be assigned to a class value or to a conditional probability. ✓
- ☐ e. The number of leaf nodes cannot be larger than twice the number of internal nodes.
- ☐ f. All decision tree learning algorithms are non-recursive.

- ☐ g. The maximal Gini impurity of a data set with M classes is $\ln M$.
- ☒ h. A split maximizes the information gain if the produced subsets are completely homogeneous. ✓

Frage 18

Richtig
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Consider a data set $\mathbf{Z}$ with 3 different class labels $y \in \{1, 2, 3\}$. There are $|\mathbf{Z}| = 1000$ data samples in the set. The probabilities p_k that a label in the data set belongs to class k are given as follows:

$$p_1 = 0.5$$

$$p_2 = 0.3$$

$$p_3 = 0.2$$

Assume that a split s perfectly splits the data set into 3 homogeneous subsets $\mathbf{Z}_{s,t}$ such that $\mathbf{Z}_{s,t}$ contains all data samples with label t ($t = 1, 2, 3$).

$$\text{Let } H \text{ denote entropy. Calculate the information gain } g_E(\mathbf{Z}, s) = H(\mathbf{Z}) - \sum_{t=1}^3 \frac{|\mathbf{Z}_{s,t}|}{|\mathbf{Z}|} H(\mathbf{Z}_{s,t}).$$

Remark: In your calculation of the entropy, use the natural logarithm $\ln = \log_e$, not a logarithm to some other base.

You can use a pocket calculator. Give the answer as a real number, rounded to two decimal places. Do not give the answer in %. Use "." for the decimal point. Number format example: 1.67

Antwort: ✓

Frage 19

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This text field is optional and will not influence the automatic grading. Here, you can write down subresults or your thoughts on the specific tasks. It may be used to your advantage at the end of the semester as mentioned in the exam info file.

$$H(\mathbf{Z}) = \text{negative sum over } p(k) * \ln(p(k)) \rightarrow -0.5 * \ln(0.5) - 0.3 * \ln(0.3) - 0.2 * \ln(0.2) = 1.0297$$

As all $\mathbf{Z}_{s,t}$ are homogeneous subsets, no further calculation is needed, as the second term in $g_E(\mathbf{Z}, s)$ is zero.

Frage 20

Richtig
Erreichte Punkte
4,00 von 4,00
Frage markieren

Which statements about ensemble methods, in particular random forests and (gradient) boosting, are true?

Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

- ☒ a. In AdaBoost, the more accurate the weak classifiers are the larger the influence they are given for the classification output. ✓
- ☐ b. Random Forests are generally harder to tune compared to Gradient Boosted Trees.
- ☐ c. Forward stagewise additive modelling is a robust boosting method which specifically depends on the quadratic loss function.
- ☒ d. A random forest has smaller variance than the individual trees if the correlation between the trees is small. ✓
- ☐ e. In Boosting, the main idea is that those samples which led to a small error (i.e. were classified well) in an iteration get a larger weight in the next iteration.
- ☐ f. In general, ensemble methods aim at reducing the bias and increasing the variance.
- ☒ g. A common measure of variable importance in random forests is the mean accuracy decrease. ✓
- ☒ h. In a random forest, each individual decision tree is built upon a random bootstrap subsample of the original data set. ✓

Frage 21

Teilweise richtig
Erreichte Punkte
3,00 von 4,00
Markierung entfernen

Which statements about AdaBoost are true?

Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

- ☒ a. In every step n , compute a misclassification error and classifier weight α_n . If classification for a data point was correct, leave corresponding weight unchanged. If classification was wrong, increase weight. ✓
- ☒ b. Final output: $g(\mathbf{x}) = \text{sign}(\sum_{n=1}^N \alpha_n g_n(\mathbf{x}))$ ✓
- ☒ c. Initialization: Weights are all initialized to the same value and sum up to 1. ✓
- ☐ d. In every step, with given weights, find optimal classifier $g_n(\mathbf{x})$ to the training data

Frage 22

Richtig
Erreichte Punkte
4,00 von 4,00
Frage markieren

Which statements about Logistic Regression (and related topics) are true?

Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

- ☒ a. In Logistic Regression, minimization of the loss function can be achieved by numerical methods such as gradient descent and conjugate gradient. ✓
- ☒ b. Assume that y data are generated around a function $g(x)$ with Gaussian noise. Minimizing the negative log-likelihood of observing the data is equivalent to minimizing the squared error between y and $g(x)$. ✓
- ☐ c. The logistic (or sigmoid) function reads: $\sigma(z) = (1 - \exp(-z))^{-1}$.
- ☒ d. Logistic regression does not have a closed-form solution. ✓
- ☐ e. The likelihood function for a Bernoulli distribution reads: $\mathcal{L}(\{\mathbf{x}, y\}; \mathbf{W}) = \prod_{i=1}^I g(\mathbf{x}_i; \mathbf{W})^{y_i} (1 - g(\mathbf{x}_i; \mathbf{W}))^{1-y_i}$.
- ☐ f. The momentum term in gradient descent is particularly helpful in steep regions of the loss function.
- ☐ g. The softmax function is a generalization of the sigmoid function and is suitable for regression.
- ☐ h. Logistic Regression is a non-convex problem.

Frage 23

Richtig
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4,00 von 4,00
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Maximum Likelihood:

Assume you have a climate model with an important parameter θ , namely the level of CO2 in the atmosphere.

You are given five data points, namely the 10-year-average global temperatures for the past 50 years.

With parameter choice θ_1 , the probabilities to obtain the given data (within a certain margin of error) are: $\{0.9, 0.9, 0.9, 0.4, 0.9\}$.

With parameter choice θ_2 , the probabilities are: $\{0.8, 0.8, 0.75, 0.75, 0.8\}$.

Enter numerical values below. You can use a pocket calculator.

- How likely is θ_1 to produce the data? Enter a real number between 0 and 1 rounded to two decimal places. Do not give the answer in %. Number format example: 0.67
- How likely is θ_2 to produce the data? Enter a real number between 0 and 1 rounded to two decimal places. Do not give the answer in %. Number format example: 0.67
- Which θ_i , $i = 1$ or 2 , is a better explanation for the data? Enter an integer number.

- $\mathcal{L}(\theta_1) =$ ✓
- $\mathcal{L}(\theta_2) =$ ✓
- $i =$ ✓

Frage 24

Vollständig

This text field is optional and will not influence the automatic grading. Here, you can write down subresults or your thoughts on the specific tasks. It may be used to your advantage at the end of the semester as mentioned in the exam info file.

Nicht bewertet

Frage
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$\theta_{1,1} = 0.9^4 \cdot 0.4 = 0.26244$
 $\theta_{1,2} = 0.8^3 \cdot 0.75^2 = 0.288$

Frage 25

Richtig

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4.00 von 4.00

Frage
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Which statements about Artificial Neural Networks are true?

Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

- ☐ a. Dropout and weight decay are regularization techniques which aim at avoiding underfitting.
- ☐ b. Let us denote the learning rate with $\eta > 0$, the weights with $\mathbf{w}$, and the empirical risk with R . Then the gradient descent update rule reads: $\mathbf{w}_{\text{new}} = \mathbf{w}_{\text{old}} + \eta \nabla_{\mathbf{w}} R$.
- ☒ c. Using the matrix-vector notation from the lecture (Unit 6), the recursive formula for delta errors is given by: $\delta^{[l-1]T} = \delta^{[l]T} \mathbf{J}^{[l]}$. Here, $\mathbf{J}^{[l]}$ is the single-layer Jacobian for layer l . ✓
- ☒ d. Multi-layer perceptrons are equivalent to fully connected feed-forward neural networks. ✓
- ☒ e. The vanishing gradient problem can be mitigated/avoided by the use of suitable activation functions. ✓
- ☐ f. The perceptron is a simple non-linear classifier.
- ☐ g. The Universal Approximation Theorem guarantees that neural networks have small VC dimension
- ☒ h. Without activation functions, neural networks are linear functions with respect to their weights. ✓

Frage 26

Teilweise richtig

Erreichte Punkte
1.33 von 4.00

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Which statements about Perceptrons and Perceptron Learning are true?

Notation as in the lecture:

Weight vector: $\mathbf{w}$

Threshold: θ

Data samples: $\mathbf{x}_i$

Learning rate: $\eta > 0$

Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

- ☐ a. If the data set is linearly separable, then there exists a stopping criterion such that with probability 1 the perceptron learning algorithm will terminate and solve the learning task.
- ☐ b. In the learning algorithm: If $g(\mathbf{x}_i; \mathbf{w}, \theta) = 1$ and $y_i = 0$, then the update rule for the threshold reads: $\theta := \theta + \eta$.
- ☐ c. The perceptron is a non-linear classifier.
- ☒ d. The final solution (values of the $\mathbf{w}$ and θ) of the perceptron learning algorithm may depend on the initialization. ✓
- ☐ e. A perceptron can solve the XOR problem, but only in certain instances, and then the solution is not unique.
- ☐ f. In the learning algorithm: If $g(\mathbf{x}_i; \mathbf{w}, \theta) = 0$ and $y_i = 1$, then the update rule for the weight vector reads: $\mathbf{w} := \mathbf{w} - \eta \mathbf{x}_i$.

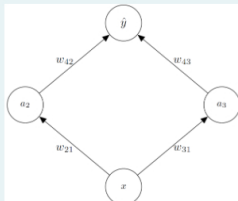
Frage 27

Richtig

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5.00 von 5.00

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Consider the following neural network:



The input unit 1 has activation $a_1 = x$. The preactivations of the hidden units 2 and 3 are denoted as s_2 and s_3 , and their activations are a_2 and a_3 , respectively. The preactivation of the output unit 4 is denoted as s_4 and the output as $a_4 = \hat{y}$. The true label is denoted as y .

In the hidden layer, we use Relu as activation function, i.e. $f_2(s) = f_3(s) = \text{Relu}(s) = \max(0, s)$. In the output layer, the activation is the identity function.

The delta-error at the output is denoted as δ_4 , and the hidden delta-errors as δ_2 and δ_3 , respectively. As loss function, we use the squared error, i.e. $L(y, \hat{y}) = \frac{1}{2}(y - \hat{y})^2$.

The learning rate is denoted as η . The network weights are updated via backpropagation and gradient descent, and these updated weights are denoted by superscript "new".

We are given the input value, the current network weights, the true label, and the learning rate for gradient descent update:

$$x = 3$$

$$w_{21} = -2$$

$$w_{31} = 2$$

$$w_{42} = -1$$

$$w_{43} = 3$$

$$y = -1$$

$$\eta = \frac{1}{100}$$

Compute the following quantities (forward pass and backpropagation):

$$\hat{y} = 18 \quad \checkmark$$

$$\delta_3 = 57 \quad \checkmark$$

$$\frac{\partial L}{\partial w_{21}} = 171 \quad \checkmark$$

$$w_{21}^{\text{new}} = 0.29 \quad \checkmark$$

You can use a pocket calculator. Give the answers as real numbers, rounded to two decimal places. Use "." for the decimal point. Number format example: 1.23

Frage 28

Vollständig

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$$s2 = -2 \cdot 3 = -6$$

$$s3 = 2 \cdot 3 = 6$$

$$a2 = 0$$

$$a3 = 6$$

```

s4 = a4 = y_hat = 0*-1 + 6*3 = 18
delta4 = y_hat - y = 18 - (-1) = 19
delta3 = relu'(s3) delta4 w_43 = 1*19*3 = 57
dL/dw_31 = delta3*a1 = 57*3 = 171
w_31^(new) = w_31^(old) - eta*171 = 2-1.71 = 0.29

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Frage 29

Richtig
Erreichte Punkte
4,00 von 4,00
Frage markieren

Convolutional Neural Networks (CNNs):

Which statements are true?

Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

- ☒ a. Dataset augmentation can help to overcome the problem of limited quantity and/or diversity of given data. ✓
- ☐ b. With kernel weight matrix $w_{i,j}$, kernel size R , and input image $x_{a,b}$, the feature map preactivation is calculated by $s_{a,b} = \sum_{i=0}^{R-1} \sum_{j=0}^{R-1} w_{i,j} x_{a,b}$.
- ☐ c. With input dimension A , kernel size R , padding P , and stride S , the feature map dimension is $(A + R + 2P)/S + 1$.
- ☒ d. Re-using a kernel weight matrix at all positions of an image is called weight sharing. ✓
- ☒ e. A convolutional layer can use multiple kernels. ✓
- ☒ f. ResNet is a famous CNN, whose good learning performance is due to skip connections. ✓
- ☐ g. Transfer Learning allows to use a kernel from one convolutional layer also in the next (neighbouring) layer.
- ☒ h. Pooling increases the receptive field through the depth of the network. ✓

Frage 30

Richtig
Erreichte Punkte
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Convolution:

You are given the input image $\mathbf{x}$ with two channels:

$$\mathbf{x}_{c=1} = \begin{pmatrix} 2 & 1 & 2 & 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 0 & 1 & 2 & 0 & 2 \\ 0 & 2 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 2 & 2 & 2 & 0 & 0 \\ 2 & 2 & 2 & 2 & 2 & 2 & 1 \\ 1 & 2 & 1 & 2 & 1 & 2 & 1 \end{pmatrix}$$

$$\mathbf{x}_{c=2} = \begin{pmatrix} 1 & 1 & 1 & 0 & 0 & 0 & 2 \\ 1 & 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 2 & 0 & 0 \\ 0 & 1 & 1 & 2 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 2 & 1 & 0 \\ 2 & 1 & 0 & 2 & 0 & 0 & 1 \\ 1 & 2 & 0 & 2 & 2 & 1 & 0 \end{pmatrix}$$

The filter $\mathbf{W}$ with depth 2 is given by

$$\mathbf{W}_{c=1} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

$$\mathbf{W}_{c=2} = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & 1 \end{pmatrix}.$$

Assuming stride equal to 1 and no padding, compute the top right entry $s_{1,6}$ of the preactivation $\mathbf{s} = \mathbf{W} * \mathbf{x}$ by convolution (cross-correlation, to be precise).

Assuming no padding and stride equal to 2, the dimension of the output matrix becomes $d \times d$. Compute d .

Assuming padding equal to 1 and stride equal to 3, the dimension of the output matrix becomes $D \times D$. Compute D .

- $s_{u,v} =$ ✓
- $d =$ ✓
- $D =$ ✓

Frage 31

Vollständig
Nicht bewertet
Frage markieren

This text field is optional and will not influence the automatic grading. Here, you can write down subresults or your thoughts on the specific tasks. It may be used to your advantage at the end of the semester as mentioned in the exam info file.

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(1 1 2) * (0 1 0) = (0)
(0 1 0) * (1 -4 1) = (-2)
(2 0 2) * (0 1 0) = (0)
+
(0 0 1) * (1 0 -1) = (-2)
(0 1 0) * (0 0 0) = (0)
(2 2 0) * (-1 0 1) = (-2)
=
(-2 -2 -2) = -6
(A - R + 2P) / S - 1
-> (7-3+0)/2+1=3
-> (7-3+2)/3+1=3

```

Frage 32

Richtig
Erreichte Punkte
4,00 von 4,00
Frage markieren

Recurrent Neural Networks (RNNs):

Which statements are true?

Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

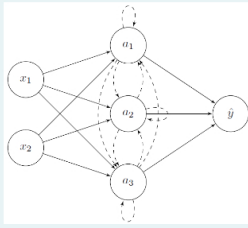
- ☐ a. In an RNN, the units in a hidden layer are all conneted with each other but cannot be connected to themselves.
- ☐ b. The constant error carousel in LSTMs must use a non-linear activation function in order to avoid the vanishing gradient problem.
- ☒ c. RNNs can be trained via Backprogration Through Time. ✓
- ☐ d. We denote with $\mathbf{R}$ the recurrent weight matrix at time t , with f the activation function, and with $\mathbf{s}(t-1)$ the preactivation at time $t-1$. For stable gradients, we need $\|\mathbf{R} \text{diag}(f(\mathbf{s}(t-1)))\| \approx 1$.
- ☐ e. Feedforward neural networks (FNNs) are Turing complete.
- ☒ f. LSTMs use an integrator to store information and gates for remembering, communicating, and (potentially) forgetting. ✓
- ☒ g. RNNs use weight sharing as the network is slid over the input sequence and the hidden-layer weight matrix is reused for all timesteps. ✓

h. RNNs are typically useful for sequence classification and generation. ✓

Frage 33

Richtig
Erreichte Punkte
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Markierung
entfernen

Consider the following recurrent neural network:



The activation function in both the hidden layer and in the output layer is Relu. The initial activation vector in the hidden layer is set to $\mathbf{a}(t=0) = (0, 0, 0)^T$.

The input vector at the first time step is $\mathbf{x}(t=1)$. The input vector at the next time step is $\mathbf{x}(t=2)$. The input weight matrix is denoted as $\mathbf{W}$. The recurrent weight matrix is denoted as $\mathbf{R}$. The output weight matrix is denoted as $\mathbf{V}$.

We are given:

$$\mathbf{x}(t=1) = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$\mathbf{x}(t=2) = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\mathbf{W} = \begin{pmatrix} 1 & 2 & 0 \\ 2 & 1 & 2 \end{pmatrix}$$

$$\mathbf{R} = \begin{pmatrix} 1 & -3 & 1 \\ -3 & 1 & 3 \\ 1 & 3 & 0 \end{pmatrix}$$

$$\mathbf{V} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$$

Compute the forward pass and calculate the network outputs $\hat{y}(t=1)$ and $\hat{y}(t=2)$:

• $\hat{y}(t=1) =$ ✓

• $\hat{y}(t=2) =$ ✓

The answers are integer numbers.

Frage 34

Vollständig

This text field is optional and will not influence the automatic grading. Here, you can write down subresults or your thoughts on the specific tasks. It may be used to your advantage at the end of the semester as mentioned in the exam info file.